NGUYEN HUU DUONG

2132300301

MIS 311

PORTFOLIO SITE

Exploratory Data Analysis (EDA)

Executive Summary

This report examines a dataset containing information from 10,000 secondary school students to determine the major factors influencing final examination results. Microsoft Excel was used to conduct the analysis, including data inspection, cleaning, descriptive statistics, PivotTable analysis, and data visualization. The findings indicate that daily study time is the most significant controllable factor affecting academic achievement. Students studying six hours or more each day achieved a perfect pass rate, whereas students studying less than one hour had extremely low success rates. In contrast, gender differences had almost no influence on academic outcomes. The analysis also shows that excessive social media use is associated with poorer performance. These results offer practical recommendations for students, parents, and educators seeking to improve learning outcomes.

1. Background & Data Overview

1.1 Context

Academic achievement remains one of the most important concerns for educational institutions. Although intelligence is commonly viewed as the key determinant of success, previous studies suggest that behavioural factors such as study habits, attendance, sleep patterns, and social media usage also strongly influence academic performance. This study aims to evaluate these relationships using a structured student dataset and to generate evidence-based insights.

1.2 Dataset Description

The dataset used in this analysis is the Student Exam Performance dataset, provided as part of the MIS 311 course materials. It captures a wide range of variables for each student, spanning demographics, home environment, study behaviour, and academic outcomes.

Attribute	**Detail**
Number of Records	10,000 students

Number of Variables	23 columns
Data Types	13 numeric (int / float), 10 categorical (text)
Key Outcome Variables	final_exam_score, pass_fail, grade_category
Key Predictor Variables	study_hours_per_day, attendance_rate, social_media_hours, previous_gpa
Student Age Range	15 to 18 years old
Gender Split	Female: 4,987 (49.9%) | Male: 5,013 (50.1%)
Data Source	MIS 311 course material, Eastern International University (EIU)

1.3 Variable Glossary (Selected)

The table below defines the most analytically relevant variables used in this study.

Variable	**Description**
study_hours_per_day	Average number of hours the student studies each day
attendance_rate	Percentage of classes attended (0–100%)
social_media_hours	Average daily hours spent on social media
assignment_completion_rate	Percentage of assignments submitted on time

previous_gpa	GPA from the previous academic period (0.0–4.0 scale)
final_exam_score	Score achieved in the final examination (0–100)
pass_fail	Binary outcome: Pass (score >= 50) or Fail (score < 50)
grade_category	Letter grade: A (80+), B (70–79), C (60–69), D (50–59), F (<50)

2. Data Cleaning

Before performing any analysis, the dataset was examined for quality issues. Two checks were conducted in Excel: (1) a blank-cell scan for missing values, and (2) a duplicate-row check on the student_id column, which serves as the unique identifier for each record.

Check	Method (Excel)	Result	Action Taken
Missing Values	Home > Find & Select > Go To Special > Blanks	0 blank cells found across all 23 columns	None required
Duplicate Rows	Data > Remove Duplicates (student_id column only)	0 duplicate student IDs detected	None required

Cleaning Outcome

The dataset is entirely complete with no missing values or duplicate records. It requires no pre-processing before analysis. This is a high-quality, ready-to-use dataset

— an uncommon characteristic that allows the analyst to proceed directly to exploration.

3. Descriptive Statistics

3.1 Summary Statistics

The following table presents key descriptive statistics for the primary numerical variables, computed in Excel using the AVERAGE, STDEV, MIN, MAX, MEDIAN, QUARTILE, and COUNT functions on each respective column.

Statistic	Age	Study Hrs/Day	Attendance %	Sleep Hrs	Soc. Media Hrs	Final Score	Prev. GPA
Count	10,000	10,000	10,000	10,000	10,000	10,000	10,000
Mean	16.49	3.02	84.70	6.98	2.15	49.68	1.98
Std Dev	1.12	1.18	8.77	0.99	1.44	12.15	0.54
Min	15	0.50	60.00	4.00	0.00	4.40	0.00
Median	16	3.01	84.70	7.00	2.20	49.55	1.99
Max	18	7.24	100.00	10.00	5.00	97.80	3.99

The average final examination score was 49.68, which is very close to the passing threshold. The median score of 49.55 suggests that the distribution is relatively balanced without severe skewness. Study hours varied significantly among students, indicating that this variable is suitable for further analysis.

3.2 PivotTable Analysis

Two PivotTables were constructed in Excel on a new sheet labelled "Analysis" to explore the relationship between student behaviour and exam outcomes.

PivotTable 1 — Academic Outcomes by Gender

Gender	Count	Avg Final Score	Avg Math	Avg Reading	Avg Attendance %	Pass Rate %
Female	4,987	49.81	49.96	50.01	84.72%	48.91%
Male	5,013	49.56	49.76	49.60	84.69%	48.25%

PivotTable 2 — Academic Outcomes by Study Hours Per Day

Study Hours	Count	Avg Final Score	Avg Math Score	Avg Attendance %	Pass Rate %
0 – 1 h	485	35.96	35.36	84.50%	6.39%
1 – 2 h	1,503	41.15	40.60	84.71%	17.50%
2 – 3 h	2,982	46.83	46.24	84.75%	36.96%
3 – 4 h	2,959	52.18	51.66	84.65%	58.70%
4 – 5 h	1,586	58.22	57.72	84.71%	80.14%

5 – 6 h	425	63.49	63.37	84.83%	92.71%
6 h +	60	69.56	69.53	84.60%	100.00%

4. Key Findings & Insights

Insight 1 — Study Hours is the Most Powerful Controllable Driver of Performance

The PivotTable data reveals a consistent, near-linear relationship between daily study hours and exam performance. Students who studied 0–1 hour per day averaged only 35.96 points with a pass rate of 6.4%. In stark contrast, students who studied 5–6 hours achieved an average of 63.49 points and a 92.7% pass rate, while the small group studying 6+ hours achieved a 100% pass rate. Every additional hour of study per day is associated with an average score increase of roughly 5–6 points. Implication: Academic support programmes should prioritise helping students establish structured, daily study routines. Counsellors and teachers should treat low study hours as an early warning indicator for academic risk.

Insight 2 — Gender Is Not a Factor, But Social Media Usage Is a Hidden Risk

Despite being a common assumption, gender shows no meaningful difference in academic performance. Female students averaged 49.81 and male students averaged 49.56 on the final exam — a gap of only 0.25 points. Pass rates are similarly close (48.91% vs. 48.25%). However, the correlation analysis reveals that social_media_hours carries a negative correlation of r = –0.246 with final_exam_score — the most harmful behavioural variable in the dataset. Combined with the fact that over 51% of all students fail, this points to a systemic, behaviour-driven performance gap rather than a demographic one. Implication: Digital wellness campaigns that help students limit social media use during study periods could meaningfully improve overall pass rates without targeting any specific demographic group.

Correlation Summary (with final_exam_score)

The table below shows the Pearson correlation coefficient between each key variable and the final exam score, as calculated in Excel using the CORREL function.

Variable	Correlation (r)	Direction

previous_gpa	+0.891	Positive
study_hours_per_day	+0.576	Positive
assignment_completion_rate	+0.171	Positive
attendance_rate	+0.151	Positive
participation_score	+0.124	Positive
sleep_hours	+0.028	Neutral
social_media_hours	–0.246	Negative

5. Visualizations

5.1 Average Final Exam Score by Gender (Bar Chart)

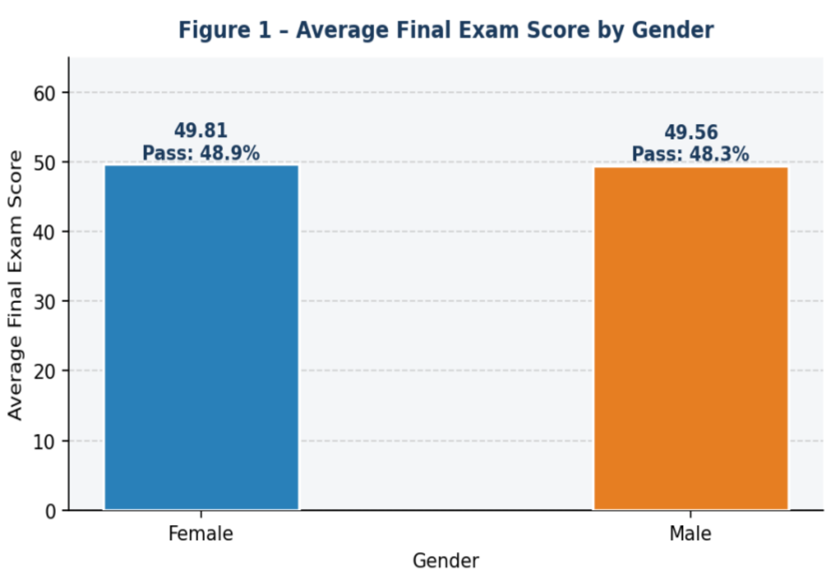

Figure 1 – Female (49.81) and male (49.56) students score almost identically. Gender is not a differentiating factor in this dataset.

5.2 Average Score & Pass Rate by Study Hours Per Day (Column + Line Chart)

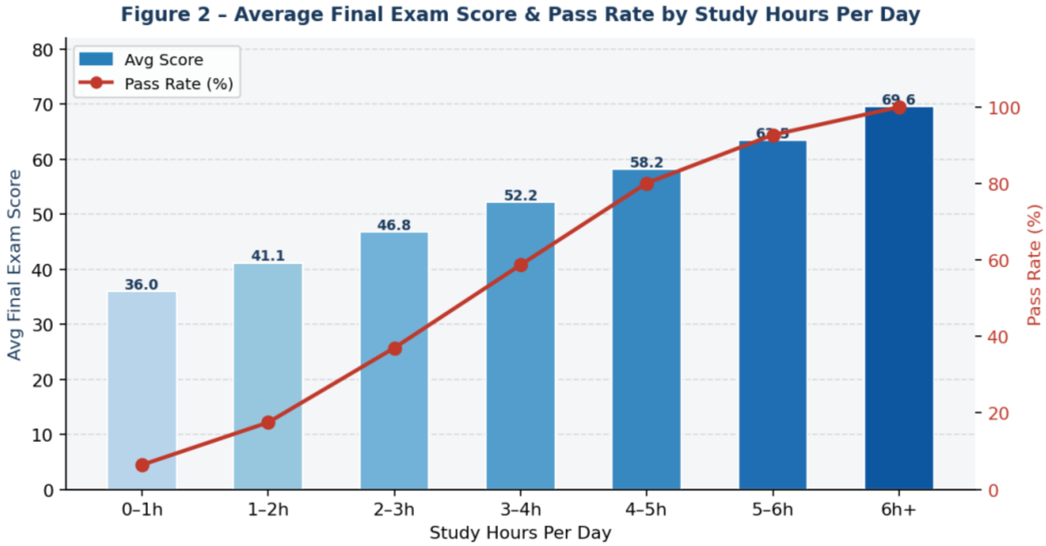

Figure 2 – A clear upward trend: every additional hour studied raises the average score by ~5 points and the pass rate by ~15 percentage points.

5.3 Distribution of Final Exam Scores (Histogram)

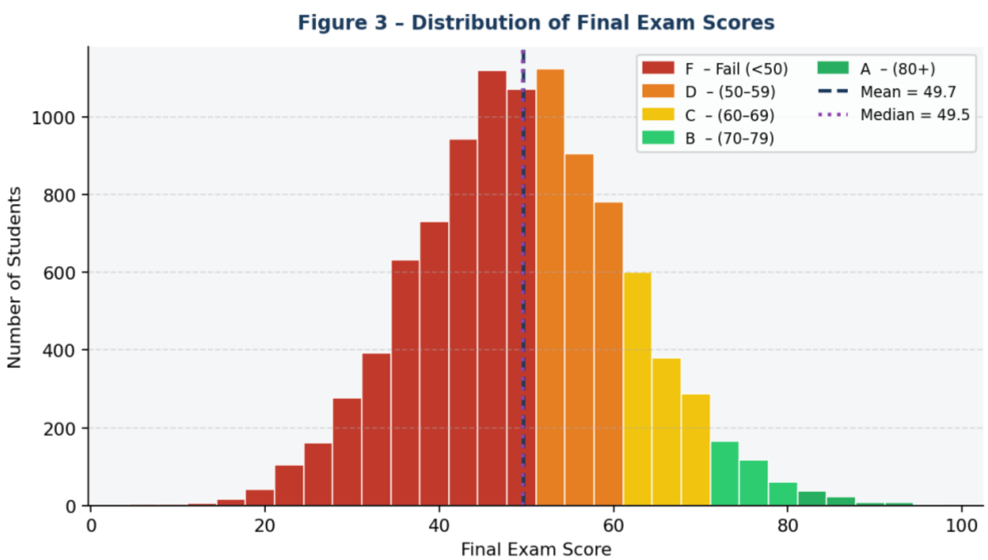

Figure 3 – The distribution is approximately normal, centred at mean = 49.7. Over half of all students fall in the red (Fail) zone, highlighting a widespread performance challenge.

6. Conclusion

This exploratory data analysis provides strong evidence that academic success is primarily influenced by student behaviour rather than demographic characteristics. Study habits, particularly the amount of daily study time, have the greatest positive impact on examination performance. Conversely, excessive social media use is associated with weaker academic outcomes.

The analysis was completed entirely in Microsoft Excel using descriptive statistics, PivotTables, correlation analysis, and charts. Since the dataset contained no missing values or duplicate records, the results can be considered reliable and accurate.

The findings suggest that educators and schools should focus on encouraging effective study routines and promoting healthier digital habits among students. Even small increases in daily study time may significantly improve both examination scores and pass rates.

References

Eastern International University (EIU). (2025). Student Exam Performance dataset [Course material]. MIS 311 – Introduction to Business Analytics.

Field, A. (2018). Discovering statistics using IBM SPSS statistics (5th ed.). SAGE Publications.

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). Multivariate data analysis (8th ed.). Cengage Learning.

Professional GitHub Portfolio

Link: https://github.com/NguyenHuuDuong-ops